

# AWS Cost Estimator

Pricing & Infrastructure Report

US

Mode: SaaS — BusinessNext  
Hosted

Date: 13 March 2026

Inflation: 4% / yr

AWS Region: us-east-1

# Executive Summary

High-level cost overview and infrastructure snapshot

|                           |                  |                    |                 |
|---------------------------|------------------|--------------------|-----------------|
| <b>\$10,398</b>           | <b>\$124,773</b> | <b>\$675,812</b>   | <b>15,500</b>   |
| Monthly Cost              | Annual Cost      | 5-Year Total       | Named Users     |
| Production + environments | Year 1 total     | At 4% inflation/yr | Year 1 licensed |

| Monthly         | Annual           | 5-Year           |
|-----------------|------------------|------------------|
| <b>\$10,398</b> | <b>\$124,773</b> | <b>\$675,812</b> |

## Project Details

|                       |                            |
|-----------------------|----------------------------|
| Customer / Bank       | US                         |
| Client Mode           | SaaS — BusinessNext Hosted |
| Database Type         | PostgreSQL                 |
| Total Named Users     | 15,500                     |
| Concurrent Users      | 4,650 (30% of named)       |
| Mobile Users          | 4,050                      |
| Worker Nodes          | 9                          |
| Total vCPUs (worker)  | 96                         |
| Total RAM GB (worker) | 172 GB                     |
| Data Size             | 2100 GB                    |
| Estimate Date         | 13 March 2026              |

# Node Distribution

AI-assisted infrastructure architecture

|                       |                       |
|-----------------------|-----------------------|
| Total Worker Nodes    | 9                     |
| Total DB Nodes        | 11                    |
| Fixed / Managed Roles | 7                     |
| Confidence            | High                  |
| Distribution Method   | Groq LLM + Rule-based |

| Category                    | Role / Service                                    | Nodes | Instance Fam.         | vCPU/node | RAM GB/node | Model         |
|-----------------------------|---------------------------------------------------|-------|-----------------------|-----------|-------------|---------------|
| Kubernetes Cluster          | Worker Nodes Linux/RHEL (Web, Mobile, WebAPI)     | 6     | Compute Intensive     | 16        | 32          | Reserved 3 Yr |
| Kubernetes Cluster          | Worker Nodes Linux/RHEL for Graphana & Prometheus | 2     | Compute Intensive     | 8         | 16          | Reserved 3 Yr |
| Kubernetes Cluster          | Worker Nodes Linux/RHEL for EFK - optional        | 1     | Compute Intensive     | 16        | 32          | Reserved 3 Yr |
| PGSQL Database              | Linux Nodes – Database server (Primary)           | 2     | Memory Intensive      | 32        | 128         | None          |
| PGSQL Database              | Linux Nodes – Cluster (etcd+haproxy, pgbackrest)  | 4     | Memory Intensive      | 2         | 8           | None          |
| PGSQL Database              | Storage: SAN (Primary DB)                         | 2     | 10K IOPS              | —         | —           | None          |
| DMS                         | DMS Storage + Replication                         | 1     | Cloud Storage Service | —         | —           | On Demand     |
| Security & Compliance       | WAF + AWS Shield Advanced                         | 1     | Managed Service       | —         | —           | On Demand     |
| Security & Compliance       | CloudTrail + Config Audit Logs                    | 1     | Managed Service       | —         | —           | On Demand     |
| Kubernetes Cluster          | Cloud Managed K8s                                 | 1     | K8s Service           | —         | —           | None          |
| Caching & Session           | Elasticaache Service                              | 2     | Memory Optimized      | 4         | 16          | Reserved 3 Yr |
| Infrastructure & Monitoring | Load Balancer (Internal + External)               | 2     | ALB                   | —         | —           | On Demand     |
| Infrastructure & Monitoring | Bastion Host                                      | 1     | General Purpose       | 2         | 4           | Reserved 3 Yr |
| Infrastructure & Monitoring | NAT Gateway                                       | 2     | NAT                   | —         | —           | On Demand     |
| Infrastructure & Monitoring | Image Registry (ECR)                              | 1     | Managed Service       | —         | —           | On Demand     |
| Infrastructure & Monitoring | Back Up - DB & Infra Logs                         | 1     | Cloud Storage Service | —         | —           | On Demand     |

# AWS Cost Breakdown

Service-by-service monthly and annual costs

## Compute & Database Roles

| Category                    | Role                                             | Nodes | Instance        | \$/hr  | \$/month | \$/year |
|-----------------------------|--------------------------------------------------|-------|-----------------|--------|----------|---------|
| Kubernetes Cluster          | Worker Nodes Linux/RHEL (Web, Mobile, WebAPI)    | 6     | m5.4xlarge      | \$0.77 | \$3,487  | \$0     |
| Kubernetes Cluster          | Worker Nodes Linux/RHEL for Grafana & Prometheus | 2     | m5.2xlarge      | \$0.38 | \$643    | \$0     |
| Kubernetes Cluster          | Worker Nodes Linux/RHEL for EFK - optional       | 1     | m5.4xlarge      | \$0.77 | \$643    | \$0     |
| PGSQL Database              | Linux Nodes – Database server (Primary)          | 2     | r5.8xlarge      | \$2.02 | \$2,991  | \$0     |
| PGSQL Database              | Linux Nodes – Cluster (etcd+haproxy, pgbackrest) | 4     | r5.large        | \$0.13 | \$400    | \$0     |
| PGSQL Database              | Storage: SAN (Primary DB)                        | 2     | EBS io2         | \$0.00 | \$525    | \$0     |
| DMS                         | DMS Storage + Replication                        | 1     | EBS io2         | \$0.00 | \$738    | \$0     |
| Kubernetes Cluster          | Cloud Managed K8s                                | 1     | EKS             | \$0.10 | \$73     | \$0     |
| Caching & Session           | ElastiCache Service                              | 2     | cache.r6g.large | \$0.33 | \$482    | \$0     |
| Infrastructure & Monitoring | Load Balancer (Internal + External)              | 2     | ALB             | \$0.00 | \$33     | \$0     |
| Infrastructure & Monitoring | Bastion Host                                     | 1     | m5.large        | \$0.10 | \$94     | \$0     |
| Infrastructure & Monitoring | NAT Gateway                                      | 2     | NAT             | \$0.00 | \$41     | \$0     |
| Infrastructure & Monitoring | Image Registry (ECR)                             | 1     | ECR             | \$0.00 | \$51     | \$0     |
| Infrastructure & Monitoring | Back Up - DB & Infra Logs                        | 1     | S3              | \$0.00 | \$118    | \$0     |
| Operations                  | CloudWatch Monitoring                            | 0     | CloudWatch      | \$0.00 | \$80     | \$0     |

## Cost Summary

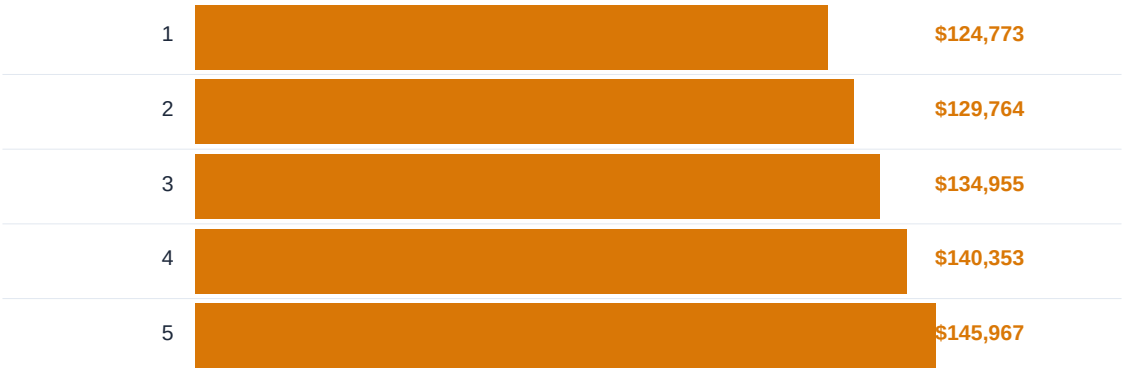
|                               |           |
|-------------------------------|-----------|
| Total Monthly (Production)    | \$10,398  |
| Total Annual (Year 1)         | \$124,773 |
| 5-Year Total (with inflation) | \$675,812 |
| Inflation Rate Applied        | 4%        |

## 5-Year Inflation Forecast

Cost projection at 4% annual inflation

| Year   | Multiplier | Monthly Cost | Annual Cost | Cumulative Total |
|--------|------------|--------------|-------------|------------------|
| Year 1 | 1.0000×    | \$10,398     | \$124,773   | \$124,773        |
| Year 2 | 1.0400×    | \$10,814     | \$129,764   | \$254,537        |
| Year 3 | 1.0816×    | \$11,246     | \$134,955   | \$389,492        |
| Year 4 | 1.1249×    | \$11,696     | \$140,353   | \$529,845        |
| Year 5 | 1.1699×    | \$12,164     | \$145,967   | \$675,812        |

### Annual Cost Trend



|                         |                                      |
|-------------------------|--------------------------------------|
| Total 5-Year Investment | \$675,812                            |
| Average Annual Cost     | \$135,162                            |
| Average Monthly Cost    | \$11,264                             |
| Year-over-Year Growth   | 4.0% (inflation) applied from Year 2 |

## Environment Pricing

Pre-Prod / SIT / UAT and DR environment costs

### Pre-Prod / SIT / UAT Environment

| Monthly        | Annual          | 5-Year     |
|----------------|-----------------|------------|
| <b>\$3,495</b> | <b>\$41,939</b> | <b>\$0</b> |

| Category       | Role                                           | Nodes | Instance        | \$/month |
|----------------|------------------------------------------------|-------|-----------------|----------|
| Kubernetes     | Cloud Managed K8s                              | 1     | EKS             | \$73     |
| Kubernetes     | Worker Nodes (Web, Mobile, WebAPI)             | 2     | m5.4xlarge      | \$1,169  |
| Kubernetes     | Worker Nodes (Grafana & Prometheus) – optional | 1     | m5.2xlarge      | \$321    |
| Kubernetes     | Worker Nodes (EFK) – optional                  | 1     | m5.4xlarge      | \$602    |
| PGSQL DB       | DB Server (Pre-Prod)                           | 1     | m5.4xlarge      | \$585    |
| PGSQL DB       | Storage: SAN Pre-Prod                          | 1     | EBS io2         | \$125    |
| PGSQL DB       | Storage: SAN SIT                               | 1     | EBS io2         | \$62     |
| PGSQL DB       | Storage: SAN UAT                               | 1     | EBS io2         | \$62     |
| DMS            | DMS Storage + Replication                      | 1     | dms.c5.large    | \$147    |
| Caching        | Caching Service (ElastiCache)                  | 1     | cache.r6g.large | \$121    |
| Infrastructure | Load Balancer (Internal + External)            | 1     | ALB             | \$16     |
| Infrastructure | NAT Gateway                                    | 2     | NAT             | \$32     |
| Infrastructure | Bastion Host                                   | 1     | m5.large        | \$94     |
| Infrastructure | Image Registry (ECR)                           | 1     | ECR             | \$50     |
| Backup         | Back Up – DB & Infra Logs                      | 1     | S3              | \$34     |

### Disaster Recovery (DR) Environment

| Monthly        | Annual           | 5-Year           |
|----------------|------------------|------------------|
| <b>\$9,504</b> | <b>\$114,052</b> | <b>\$642,454</b> |

| Category   | Role                                    | Nodes | Instance   | \$/month |
|------------|-----------------------------------------|-------|------------|----------|
| Kubernetes | Cloud Managed K8s (DR)                  | 1     | EKS        | \$73     |
| Kubernetes | Worker Nodes (Web, Mobile, WebAPI) – DR | 4     | m5.4xlarge | \$2,324  |
| Kubernetes | Worker Nodes (Grafana) – DR             | 2     | m5.2xlarge | \$643    |
| Kubernetes | Worker Nodes (EFK) – DR                 | 2     | m5.2xlarge | \$724    |
| PGSQL DB   | DB Server Primary – DR                  | 2     | m5.8xlarge | \$2,291  |
| PGSQL DB   | DB Cluster (etcd+haproxy) – DR          | 4     | m5.large   | \$312    |
| PGSQL DB   | Storage: SAN Primary – DR               | 2     | EBS io2    | \$750    |

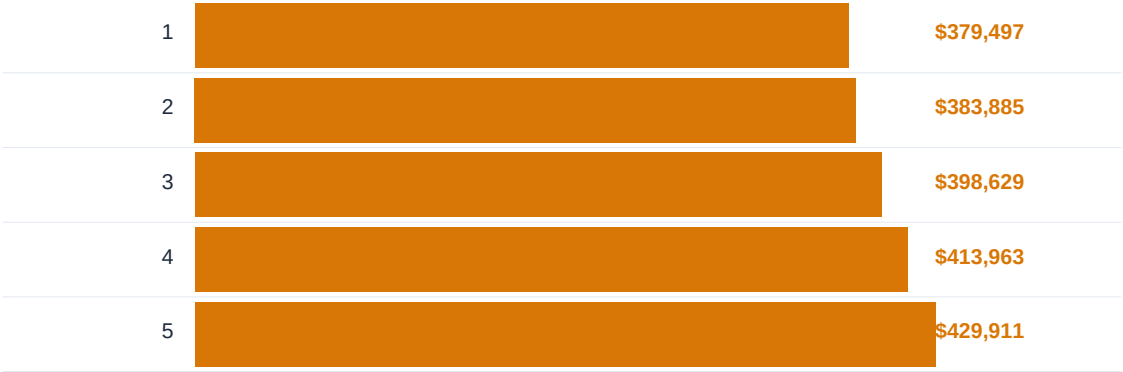
| Category       | Role                           | Nodes | Instance        | \$/month |
|----------------|--------------------------------|-------|-----------------|----------|
| PGSQL DB       | Reporting DB – DR              | 1     | m5.8xlarge      | \$1,145  |
| PGSQL DB       | Storage: SAN Reporting – DR    | 1     | EBS io2         | \$375    |
| DMS            | DMS Storage + Replication – DR | 1     | dms.c5.large    | \$296    |
| Caching        | Elastichache Service – DR      | 2     | cache.r6g.large | \$242    |
| Infrastructure | Load Balancer – DR             | 2     | ALB             | \$33     |
| Infrastructure | NAT Gateway – DR               | 2     | NAT             | \$32     |
| Infrastructure | Bastion Host – DR              | 1     | m5.large        | \$94     |
| Infrastructure | Image Registry (ECR) – DR      | 1     | ECR             | \$51     |
| Backup         | Back Up – DR                   | 1     | S3              | \$118    |

# PUPM Analysis

Price Per User Per Month — 5-year projection

| Year   | Named Users | Monthly Usage | Platform Annual | AWS Total | Discounted Cost | PUPM     |
|--------|-------------|---------------|-----------------|-----------|-----------------|----------|
| Year 1 | 15,500      | \$24,405      | \$304,323       | \$319,540 | \$379,497       | \$2.0403 |
| Year 2 | 16,275      | \$25,341      | \$315,993       | \$331,793 | \$383,885       | \$1.9656 |
| Year 3 | 17,088      | \$26,315      | \$328,130       | \$344,537 | \$398,629       | \$1.9440 |
| Year 4 | 17,943      | \$27,327      | \$340,753       | \$357,790 | \$413,963       | \$1.9226 |
| Year 5 | 18,840      | \$28,380      | \$353,880       | \$371,574 | \$429,911       | \$1.9016 |

## Discounted Annual Cost Trend



## Formula Applied

|                     |                                                                    |
|---------------------|--------------------------------------------------------------------|
| Business Support    | 3.91% of Total Monthly Usage                                       |
| Buffer              | 5% of Total Platform Annual Cost                                   |
| Managed Services    | 30% of Total AWS Annual Cost                                       |
| Discount            | 11% off Total Cost                                                 |
| PUPM                | Discounted Annual Cost ÷ 12 ÷ Named Users                          |
| User Growth         | +5% per year (Year-over-Year)                                      |
| One-Time Costs (Y1) | Performance: \$5,000 · Migration: \$5,000 · Managed Setup: \$1,000 |



# GCP Compute Engine Pricing

Google Cloud equivalent — US Central (Iowa)

| Monthly  | Annual    | 5-Year      |
|----------|-----------|-------------|
| \$16,141 | \$193,688 | \$1,049,076 |

| Category                    | Role                                             | Nodes | GCP Machine                    | \$/hr  | \$/month | \$/year  |
|-----------------------------|--------------------------------------------------|-------|--------------------------------|--------|----------|----------|
| Kubernetes Cluster          | Worker Nodes Linux/RHEL (Web, Mobile, WebAPI)    | 6     | n2-standard-16                 | \$1.17 | \$5,378  | \$64,540 |
| Kubernetes Cluster          | Worker Nodes Linux/RHEL for Grafana & Prometheus | 2     | n2-standard-8                  | \$0.58 | \$1,027  | \$12,323 |
| Kubernetes Cluster          | Worker Nodes Linux/RHEL for EFK - optional       | 1     | n2-standard-16                 | \$1.17 | \$1,027  | \$12,323 |
| PGSQL Database              | Linux Nodes – Database server (Primary)          | 2     | n2-highmem-32                  | \$3.86 | \$5,734  | \$68,804 |
| PGSQL Database              | Linux Nodes – Cluster (etcd+haproxy, pgbackrest) | 4     | n2-highmem-2                   | \$0.24 | \$772    | \$9,263  |
| PGSQL Database              | Storage: SAN (Primary DB)                        | 2     | Persistent Disk Extreme        | \$0.00 | \$525    | \$6,300  |
| DMS                         | DMS Storage + Replication                        | 1     | Persistent Disk Extreme        | \$0.00 | \$738    | \$8,850  |
| Kubernetes Cluster          | Cloud Managed K8s                                | 1     | GKE Standard Cluster           | \$0.10 | \$73     | \$876    |
| Caching & Session           | Elasticache Service                              | 2     | Memorystore Redis (redis-16gb) | \$0.24 | \$358    | \$4,292  |
| Infrastructure & Monitoring | Load Balancer (Internal + External)              | 2     | Cloud Load Balancing           | \$0.00 | \$35     | \$420    |
| Infrastructure & Monitoring | Bastion Host                                     | 1     | n2-standard-2                  | \$0.15 | \$107    | \$1,279  |
| Infrastructure & Monitoring | NAT Gateway                                      | 2     | Cloud NAT                      | \$0.00 | \$32     | \$384    |
| Infrastructure & Monitoring | Image Registry (ECR)                             | 1     | Artifact Registry              | \$0.00 | \$51     | \$614    |
| Infrastructure & Monitoring | Back Up - DB & Infra Logs                        | 1     | Cloud Storage (Nearline)       | \$0.00 | \$205    | \$2,458  |
| Operations                  | Cloud Operations (Logging + Monitoring)          | 0     | Cloud Operations Suite         | \$0.00 | \$80     | \$960    |

## GCP 5-Year Forecast

| Year   | Multiplier | Monthly Cost | Annual Cost | Cumulative  |
|--------|------------|--------------|-------------|-------------|
| Year 1 | 1.0000×    | \$16,141     | \$193,688   | \$193,688   |
| Year 2 | 1.0400×    | \$16,786     | \$201,435   | \$395,123   |
| Year 3 | 1.0816×    | \$17,458     | \$209,493   | \$604,616   |
| Year 4 | 1.1249×    | \$18,156     | \$217,872   | \$822,488   |
| Year 5 | 1.1699×    | \$18,882     | \$226,587   | \$1,049,076 |

# AWS vs GCP Cost Comparison

Side-by-side cost analysis — same workload, two clouds

|                 |                 |                |                 |
|-----------------|-----------------|----------------|-----------------|
| <b>\$10,398</b> | <b>\$16,141</b> | <b>\$5,743</b> | <b>AWS</b>      |
| AWS Monthly     | GCP Monthly     | Monthly Diff   | 5-Yr Winner     |
| us-east-1       | us-central1     | AWS is cheaper | Saves \$373,264 |

|                          |             |
|--------------------------|-------------|
| <b>AWS Annual</b>        | \$124,773   |
| <b>GCP Annual</b>        | \$193,688   |
| <b>AWS 5-Year Total</b>  | \$675,812   |
| <b>GCP 5-Year Total</b>  | \$1,049,076 |
| <b>Cheaper (Monthly)</b> | AWS         |
| <b>Cheaper (5-Year)</b>  | AWS         |
| <b>5-Year Saving</b>     | \$373,264   |

## Category-Level Breakdown

| Category                    | AWS \$/mo | GCP \$/mo | Difference | % Diff | ✓ Cheaper |
|-----------------------------|-----------|-----------|------------|--------|-----------|
| Caching & Session           | \$482     | \$358     | \$124      | 34.7%  | GCP       |
| DMS                         | \$738     | \$738     | \$0        | 0.0%   | AWS       |
| Infrastructure & Monitoring | \$337     | \$430     | \$92       | 21.5%  | AWS       |
| Kubernetes Cluster          | \$4,845   | \$7,505   | \$2,660    | 35.4%  | AWS       |
| Operations                  | \$80      | \$80      | \$0        | 0.0%   | AWS       |
| PGSQL Database              | \$3,916   | \$7,031   | \$3,114    | 44.3%  | AWS       |
| Security & Compliance       | \$0       | \$0       | \$0        | 0.0%   | AWS       |

## 5-Year Year-by-Year Comparison

| Year   | AWS/mo   | GCP/mo   | AWS/yr    | GCP/yr    | AWS Cumul. | GCP Cumul.  | Winner |
|--------|----------|----------|-----------|-----------|------------|-------------|--------|
| Year 1 | \$10,398 | \$16,141 | \$124,773 | \$193,688 | \$124,773  | \$193,688   | AWS    |
| Year 2 | \$10,814 | \$16,786 | \$129,764 | \$201,435 | \$254,537  | \$395,123   | AWS    |
| Year 3 | \$11,246 | \$17,458 | \$134,955 | \$209,493 | \$389,492  | \$604,616   | AWS    |
| Year 4 | \$11,696 | \$18,156 | \$140,353 | \$217,872 | \$529,845  | \$822,488   | AWS    |
| Year 5 | \$12,164 | \$18,882 | \$145,967 | \$226,587 | \$675,812  | \$1,049,076 | AWS    |

# Notes & Assumptions

|                   |                                                                                                                                                                  |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AWS Pricing       | All prices are AWS On-Demand, us-east-1 region, Linux/UNIX, shared tenancy. Reserved or Savings Plans pricing will reduce costs by 30–40%.                       |
| Inflation Rate    | 4% per year applied to all compute costs from Year 2 onwards.                                                                                                    |
| Fallback Rates    | If live AWS Pricing API is unavailable, hardcoded March 2026 estimates are used.                                                                                 |
| Node Distribution | Distribution computed via rule-based algorithm enhanced by Groq LLM (llama-3.3-70b-versatile) at temperature=0. SHA-256 cache prevents redundant calls.          |
| Security Costs    | Airtel SOC (\$500/mo), SOC Machines (Rs 30,000/yr), Req 4 Antivirus (Rs 2,600/yr), Req 6 Data Discovery (Rs 4,000/yr) are fixed INR-converted estimates.         |
| Pre-Prod / DR     | Pricing derived from reference sizing workbook. PostgreSQL uses self-hosted EC2 (Patroni HA). SQL Server and Oracle use AWS RDS Managed.                         |
| Named Users       | Year-over-year growth assumed at 5%. Concurrent users assumed at 30% of named users.                                                                             |
| GCP Pricing       | GCP prices are estimates using hardcoded us-central1 On-Demand rates (March 2026). GCP Cloud Billing API integration available via GOOGLE_CLOUD_PROJECT env var. |
| Confidentiality   | This document contains confidential cost estimates. Do not distribute outside the project team without authorisation.                                            |

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